

Architecting for Action: Data Lineage and Analysis via the T-DAR Framework

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Abstract

Data lineage is a critical aspect of modern data management, providing insights into the flow and transformation of data across complex ecosystems. This paper introduces the T-DAR framework, a novel approach to analyzing data lineage that emphasizes the relationship between targeted observations, data assets, and actionable results. By leveraging the T-DAR framework, organizations can enhance their understanding of data processes, improve decision-making, and ensure compliance with regulatory requirements.

1 Introduction

In the era of Big Data, organizations have adopted a data-driven approach to decision-making. However, the complexity of data ecosystems often leads to challenges in understanding the flow and transformation of data. This paper introduces the **T-DAR framework**, which stands for Targeted Data-Actions-Results, as a methodology for analyzing data lineage and ensuring actionable insights. The T-DAR framework provides a structured approach to capturing and analyzing the relationships between data assets, the actions taken on them, and the resulting outcomes. By focusing on targeted observations, organizations can gain a deeper understanding of their data processes and make informed decisions based on actionable insights.

2 T-DAR Framework Overview

The T-DAR framework is inspired by the Data-Action-Result (DAR) model of most healthcare information systems.

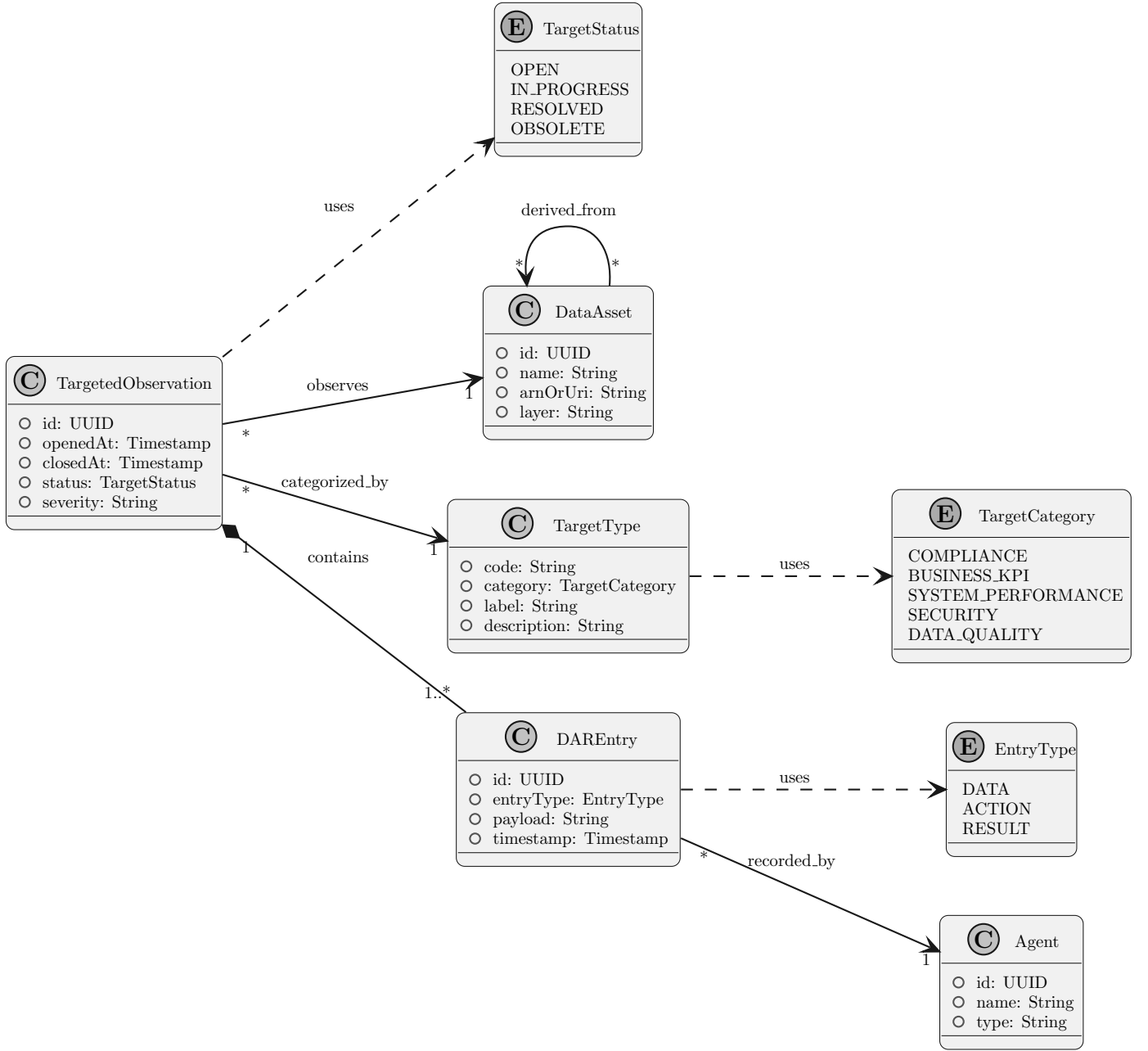


Figure 1: T-DAR Architecture Overview

As illustrated in Figure 1, the architecture of the T-DAR framework is centered around a *Targeted Observation*, which is directly linked to a specific *Data Asset*

within the ecosystem. Rather than treating data lineage as a static representation of data streams, the T-DAR framework takes advantage of the dynamic nature of data environments, allowing for real-time analysis and decision-making. It provides a way to analyse not only results, but also the medium through which the results are obtained, and the actions that lead to those results.

To operationalize this dynamic analysis, every *Targeted Observation* acts as a container for a chronological sequence of *DAREntry* records. These entries are categorized into three distinct phases: *Data*, *Action*, and *Result*, which together form the foundation of the framework:

- **Data (D) – The Assessment:** A Data entry represents a factual snapshot, an audit trail, or a metric collected from the *Data Asset*.
- **Action (A) – The Intervention:** An Action entry represents a specific action taken in response to the data or the results of the data.
- **Result (R) – The Outcome:** A Result entry represents the outcome or consequence of the preceding action taken.

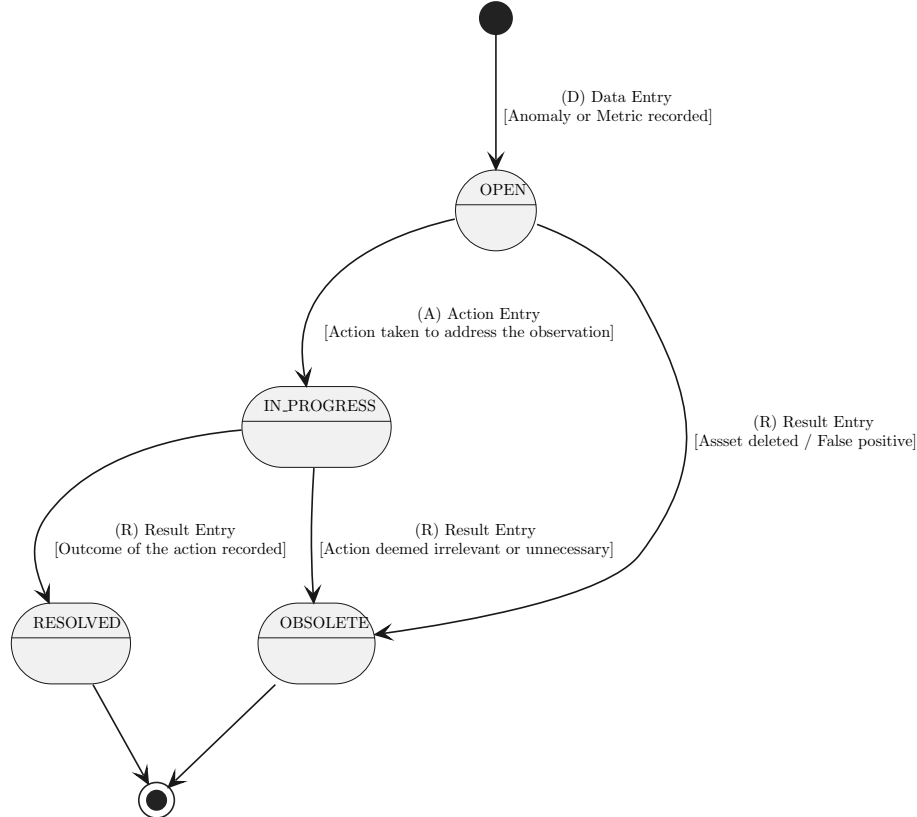


Figure 2: T-DAR State Diagram

As shown in Figure 2, these DAR entries act as triggers that drive the lifecycle of a *Targeted Observation*, transitioning it from an initial open state to a final resolution.

Furthermore, since data assets are inherently interconnected through a **derived** **from** lineage relationship (as modeled in the framework architecture), the T-DAR framework allows organizations to track how an *Action* applied to an upstream asset propagates its *Results* to downstream dependencies. This elevates traditional data lineage from a simple dependency graph to an actionable governance tool.